

Week 1.9 — Election Day: The Team Becomes a Company, and the ROV Gets a Brain

Sunday, August 16, 2026 · 3:00 – 5:00 PM · 13 students

Class 1.9 is the day Say Watt Robotics stopped being a group of kids who build a robot and became a company. First we finished what we decided at 1.8 and actually registered where our documents live: GitHub for code, Google Drive for everything else, with everybody signed up. Then came the demo that made the digital side click — a mock remote operated vehicle running on an Arduino microcontroller with a real graphical user interface, built across the Arduino IDE and VS Code. Seeing a joystick move something on a screen through code we could read is a different thing than reading about it. After that we talked about how real engineering organizations work: the integrated product team, where the leads have to talk to each other instead of building in parallel and colliding at the end, and the integrated master schedule, the one calendar that says what every sub-team owes and when. Then we ran the elections. We went back through everyone's stated position preferences from 1.8 and seated the three executives, and the executives turned around and assigned the lead roles themselves. Seventeen positions, every one filled, every one a real job with real duties. Nobody on this team is a spare part.

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POSITIONS FILLED	EXECUTIVES SEATED, WHO PICKED THE LEADS	STUDENTS AT THE SESSION	SYSTEMS OF RECORD — GITHUB AND DRIVE
THE COMPANY — ELECTED 16 AUGUST 2026			
CEO	Trenton	Sensor Lead	Rhylan
CFO	Maddox	Tether Lead	Rebecca
COO	Annabelle	Pilot Station / Control Lead	Tyler
CTO / Chief Engineer	Isaac	Software / Telemetry Lead	Oliver
Mechanical Lead	Adrian	Test & Quality / Safety & Compliance Lead	Frankie
Propulsion Lead	Sam	Marketing / Communications / Outreach	Lillian
Payload / Manipulator Lead	Eli	Document Manager Lead	Luke
Buoyancy / Stability Control Lead	Evan	Strategic Lead	Carter
Electrical / Power Lead	Liam		

The three executive officers were seated first, from the position preferences every student wrote down at 1.8. The executives then assigned the lead roles themselves — which is how it works in a real company, and which means the leads answer to teammates, not to a coach. Full duty lists for every position are on the Team page at saywattrobotics.org.

WHAT WE LEARNED

- An integrated product team is not a seating chart. The point is that the person designing the frame and the person designing the payload talk while there's still time to change something — not at integration, when the only option left is a workaround.
- An integrated master schedule turns "we should probably do that soon" into a date somebody owns. One calendar, every sub-team, every deliverable.
- A microcontroller with a GUI is the bridge between what we built and what we need. The demo showed the whole chain — joystick input, code we can read, something moving in response.
- Picking leaders from what people actually want to do beats assigning by who's loudest. Everyone wrote preferences a week before anyone got a title.
- Executives who pick their own leads own the outcome differently than executives who inherit them.

DECISIONS MADE

- GitHub for code, Google Drive for everything else — registered, not just agreed to. Everybody has an account and knows which system a file belongs in.
- The three executives were seated from the students' own stated preferences, and then the executives assigned the leads. Coaches did not hand out titles.
- Every position is a real job with a written duty list. A title nobody can describe the work of is not a title.
- The team runs as an integrated product team — leads talk during the week, not at integration.
- The COO owns an integrated master schedule: one calendar, every deliverable and date on it.

ACTION ITEMS

Publish the integrated master schedule — every sub-team's deliverables and due dates on one calendar

Annabelle (COO) · by 1.10

Run the first integrated product team meeting with all leads reporting status

Trenton (CEO) · by 1.10

Confirm every student has GitHub and Google Drive access, and file the first documents in the right place

Luke (Document Manager) · by 1.10

Each lead reads their duty list and comes back with the first three things they own this season

All leads · by 1.10

Extend the Arduino mock-ROV demo toward the real vehicle — joystick to microcontroller to motor driver

Oliver (Software) · by 1.10

Carry over from 1.8: tunable buoyancy design, motor-placement trade study, and the Sabertooth swap

Adrian / Evan / Liam · by 1.10

STUDENT LEARNING OUTCOMES

- Every student now holds a named position with a written duty list — seventeen real jobs, no spare parts.
- Saw a complete digital control chain end to end: joystick input, Arduino microcontroller, code in the IDE, and a graphical interface responding in real time.
- First exposure to two professional development environments — the Arduino IDE and VS Code — and why a team might use both.
- Learned what an integrated product team and an integrated master schedule are, and why organizations that skip them fail at integration.
- Registered on the team's real systems of record: GitHub for code, Google Drive for documents.
- Practiced how a real organization seats leadership — preferences first, executives seated from them, executives assigning their own leads.

LOOKING AHEAD

Class 1.10 is the first week the org chart has to do actual work. The COO publishes the integrated master schedule, the CEO runs the first integrated product team meeting, and every lead comes back with the first three things they own. The engineering carryover from 1.8 is still live: tunable buoyancy, the motor-placement trade study, and the Sabertooth swap. And the software track keeps pushing the Arduino demo toward the real vehicle.

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